Towards Scalable and Generalizable Video Retrieval

Supplementary Material

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APPENDIX A

SUPPLEMENTARY MATERIAL FOR TOWARDS UNIVERSAL VIDEO RETRIEVAL: GENERALIZING VIDEO EMBEDDING VIA SYNTHESIZED MULTIMODAL PYRAMID CURRICULUM

A.1 Overview of Appendix

This appendix provides supplementary material supporting the main chapter, organized as follows:

Methodology Supplementary.

- **Details of Evaluations (for Section 3.2):** (1) *UVRB Details (Appendix A.1.1):* Provides statistics, construction strategies, and evaluation details for all 16 datasets in UVRB. (2) *Evaluation Pipeline (Appendix A.1.2):* Offers a technical overview of the scalable and reproducible evaluation protocol.

- **Details of Data Synthesis and Preparation (for Section 3.3.1):** (1) *Training Data (Appendix A.1.3):* Details the composition and scale of the multi-modal training mixture. (2) *Dataset Construction Pipeline (Appendix A.1.4):* Explains the design principles of the unified dataset framework. (3) *Synthetic Data Prompts (Appendix A.1.5):* Presents the structured prompts used for generating synthetic video retrieval data.

- **Details of Model and Training (for Section 3.3.2):** (1) *Model Architecture (Appendix A.1.6):* Gives a detailed description of the GVE model architecture,

including input fusion and embedding extraction. (2) *Training Implementation (Appendix A.1.7):* Outlines the parameter-efficient fine-tuning strategy and key hyperparameters.

Experimental Details and Results.

- **Baseline Details (Appendix A.1.8):** Lists the specifications of all 14 evaluated baseline models.
- **Training Dynamics (Appendix A.1.9):** Visualizes training dynamics and metrics for the GVE-3B and GVE-7B models.
- **Additional Experiments**:
  (1) **Training-time Scaling (Appendix A.1.10):** Presents the impact of scaling training data on performance, complementing the analysis in the main chapter.
  (2) **Test-time Scaling (Appendix A.1.11):** Analyzes the effect of scaling test-time parameters (e.g., number of frames, resolution) on model performance.
  (3) **Data Quality Validation (Appendix A.1.12):** Details the human-in-the-loop evaluation conducted to validate the quality of our synthesized UVRD dataset.
  (4) **Curriculum Analysis (Appendix A.1.13):** Provides in-depth experimental analysis to validate the design and effectiveness of the Modality Pyramid.
  (5) **Video Classification (Appendix A.1.14):** Reports zero-shot classification performance on standard video action recognition benchmarks.

A.1.1 UVRB Details

This section provides details of UVRB as shown in Figure A.1, including statistics and construction strategies of datasets.

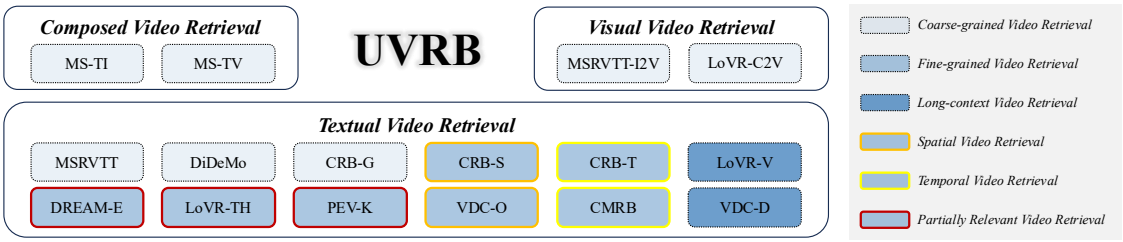


Figure A.1: The dataset category of Universal Video Retrieval Benchmark (UVRB).

First, the statistics for the number of queries and corpus, the video durations, and the text lengths are presented in Table A.1.

Second, we introduce the construction strategies of each dataset as follows.

- **MSRVTT:** We follow the data partition from JSFusion [10], utilizing 1,000 clip-text pairs from the original MSRVTT dataset [1] for evaluation.
- **DiDeMo:** Following the methodology in [11], we concatenate all sentence descriptions associated with a single video from the DiDeMo dataset [2] to create a paragraph-level query for paragraph-to-video retrieval.
- **CRB-G:** We adhere to the CaReBench protocol [3] and use the content from the `caption` field as the general query to retrieve videos.
- **CRB-S:** Similar to CRB-G, we follow [3] and select the text from the `spatial_caption` field to form queries focused on spatial descriptions.
- **VDC-O:** We utilize the VDC dataset [4] and extract text from the `main_object_caption` field as an object-centric query (e.g., *The main subject, a worker dressed in a gray sleeveless shirt and beige pants...*).
- **CRB-T:** Similar to CRB-S, we again follow [3], using the `temporal_caption` field to create queries based on temporal progression.

Table A.1: Statistics of datasets in the Universal Video Retrieval Benchmark (UVRB). All videos use 8 uniformly sampled frames. # Query: the number of queries; # Corpus: the number of corpus; Dur (s): Duration in seconds; # Word: text length in words.

Dataset	**# Query**	**# Corpus**	**Dur (s)**	**# Word**
Textual Video Retrieval (Coarse-grained)				
MSRVTT [1]	1,000	1,000	15.0	9.4
DiDeMo [2]	1,004	1,004	53.9	29.1
CaReBench-General (CRB-G) [3]	1,000	1,000	14.4	232.2
Textual Video Retrieval (Fine-grained)				
(a) Spatial				
CaReBench-Spatial (CRB-S) [3]	1,000	1,000	14.4	115.0
VDC-Object (VDC-O) [4]	1,027	1,027	30.1	91.4
(b) Temporal				
CaReBench-Temporal (CRB-T) [3]	1,000	1,000	14.4	103.2
CameraBench (CMRB) [5]	728	1,071	5.7	24.8
(c) Partially Relevant				
DREAM-1K-Event (DREAM-E) [6]	6,251	1,000	8.8	6.5
LoVR-Theme2Clip (LoVR-TH) [7]	8,854	8,854	16.9	48.1
PE-Video-Keyword (PEV-K) [8]	14,427	15,000	16.9	45.5
Textual Video Retrieval (Long-context)				
LoVR-Text2Video (LoVR-V) [7]	100	467	1560.3	17364.5
VDC-Detail (VDC-D) [4]	1,000	1,027	30.1	508.0
Composed Video Retrieval				
MomentSeeker-Text-Image (MS-TI) [9]	400	10	13.5	68.5
MomentSeeker-Text-Video (MS-TV) [9]	400	10	13.5	68.5
Visual Video Retrieval				
MSRVTT-ImageVideo (MSRVTT-I2V) [1]	1,000	1,000	15.0	-
LoVR-Clip-to-Video (LoVR-C2V) [7]	467	467	1560.3	-

- **CMRB:** We use the detailed camera motion annotations from CameraBench [5] as queries to retrieve videos (e.g., *The camera smoothly dollies forward, maintaining a steady and fluid motion...*).

- **DREAM-E:** For event-based retrieval, we collect event descriptions from the DREAM-1K dataset [6] to serve as queries for event-to-video matching (e.g., *Wooden trap launches purple squirrel into the air*).

- **LoVR-TH:** From the LoVR dataset [7], we select the theme annotations of video clips as queries for theme-to-clip retrieval (e.g., *The overall style of the animation is vibrant and whimsical...*).

- **PEV-K:** We use the annotations from the `keyword` field in the PE-Video test set [8] to perform keyword-based partially relevant matching (e.g., *colorful, paper, beautiful...*).

- **LoVR-V:** We leverage the full-length video captions from the LoVR dataset [7] to perform long-text to long-video retrieval.

- **VDC-D:** For this task, we extract the long, detailed descriptions from the `detailed_caption` field of the VDC dataset [4] to serve as fine-grained queries.

- **MS-TI:** Following the adaptation method in [12] for the MomentSeeker dataset [9], we create a text-image composed retrieval task. The goal is to retrieve a target video clip using a combined text and image query.

- **MS-TV:** Similar to MS-TI, the task for MS-TV is to use a composed query of a text description and a reference video clip to retrieve the target clip.

- **MSRVTT-I2V:** We construct image-to-video retrieval pairs from the aforementioned MSRVTT test set. For each video, a single frame is randomly sampled to serve as the image query for retrieving its source video.

- **LoVR-C2V:** We leverage the original clip-to-video structure of the LoVR dataset [7]. For each full-length video, one of its corresponding short clips is used as the query for clip-to-video retrieval.

Third, we list the query prompt and metrics for datasets in evaluations in Table A.2.

Table A.2: Query Prompts and Metrics for Datasets in UVRB.

Dataset	Query Prompt	Metric
MSRVTT	Find the clip that corresponds to the described scene in the given video.	Recall@1
DiDeMo	Find a video that includes the following described scenes.	Recall@1
CRB-G	Find the video according to the general text description.	Recall@1
CRB-S	Find the video according to the spatial description.	Recall@1
VDC-O	Find the video according to the object description.	Recall@1
CRB-T	Find the video according to the temporal description.	Recall@1
CMRB	Find the video according to the camera motion description.	Recall@10
DREAM-E	Find the video according to the text description.	Recall@1
LoVR-TH	Find the video according to text description about video theme information.	Recall@10
PEV-K	Find the video according to the text description of a series of keywords.	Recall@1
LoVR-V	Find the long video according to the long text description.	Recall@1
VDC-D	Find the video according to the detailed text description.	Recall@1
MS-TI	Find the video clip that corresponds to the given text and the given image.	Precision@1
MS-TV	Find the video clip that corresponds to the given text and the given video.	Precision@1
MSRVTT-I2V	Find the video according to the image.	Recall@1
LoVR-C2V	Find the original long video according to the short video clip.	Recall@1

A.1.2 Evaluation Pipeline

Our evaluation is built atop MTEB [13]. It decouples the evaluation engine from model architecture—supporting everything from sentence transformers to custom multimodal

encoders—via a standardized interface. The pipeline operates in three phases: (1) dynamic task orchestration, (2) configurable model execution, and (3) generation of metrics and diagnostics. For instruction-sensitive tasks, it dynamically injects domain-specific prompts to mirror real-world conditions for actionable diagnostics. Custom benchmarks are integrated seamlessly through dedicated loaders. In this framework, expensive tasks are deferred for efficiency, model initialization (precision, multi-GPU) is abstracted via a factory, and data loading supports both online and offline modes for robustness across environments. Engineering safeguards, including explicit multi-GPU management, clean shutdowns, and offline, first retry logic—ensure reliability at scale.

A.1.3 Training Data

To build a robust and versatile multimodal retrieval system, our models are trained on a large-scale, diverse mixture of text, image, and video data. All tasks are uniformly formulated as instruction-guided retrieval, a strategy designed to foster a unified representation space that can adeptly handle a wide array of queries. The training data is organized into two primary collections: a main mixture of widely-used public datasets (Table A.3), and a synthesized set from our Universal Video Retrieval Dataset (UVRD) suite (Table A.4).

We follow existing works [14, 15] to construct text-only (e.g., MSMARCO) and image-centric data (e.g., CIRR). In addition, we prepare video datasets using the following strategies.

- **VAST** [16]: We randomly select one sentence from the multiple vision captions as the text query for each video.
- **InternVid-FLT** [17]: We drop about 300K low-quality videos and use the left text-video pairs.
- **PE-Video** [8]: We choose the human_caption refined on model_caption as the textual

description of videos.

- **WebVid** [18]: We refuse the queries captioning over one video to impede the generality of text, which excludes 5M videos in final training approximately.

As the aforementioned methodology, we train our model based on contrastive learning with pre-mined explicit hard negatives and in-batch negatives. To maintain a balanced training diet and manage computational load, we employ a sophisticated data sampling strategy. For extremely large datasets, we sample a fixed number of instances. To amplify the learning signal from certain high-value or complex datasets, we may apply an upsampling strategy by repeating their data.

The scale of our training data is substantial. For our 3B parameter model, we prepared 12.55 million instances. To further leverage the capacity of our 7B parameter model, we increase the sampling rate for several large-scale datasets, bringing the total instances to 13.73 million. A detailed breakdown of the data composition is provided in Table A.5.

A.1.4 Dataset Construction Pipeline Details

Current multimodal retrieval research is hindered by fragmented, ad-hoc dataset construction. We introduce a unified, object-oriented framework that elevates this process to a rigorous science, balancing conceptual clarity with robustness for scalable, reproducible benchmarking.

The framework is anchored in a canonical tripartite abstraction: every retrieval task is decomposed into a Corpus, a set of Queries, and a Relevance Mapping. This schema, enforced by an abstract base class, is a conceptual invariant that ensures structural consistency across all datasets, from text-to-video to complex composed queries, enabling seam-

Table A.3: Configuration of the main training data mixture for our 3B and 7B models. 'K' denotes thousands, 'M' denotes millions. Notes: (a) Task: T=Text, I=Image, V=Video. The format is Query→Corpus. (b) Neg.: Number of explicit hard negatives per positive. '0' indicates use of in-batch negatives only.

Dataset	Task	Sample Size		Neg.	BS
		3B	7B		
Part 1: Text-only Data					
MSMARCO	T→T	300K	500K	2	64
HotpotQA	T→T	69K		1	64
WebQA	T→T	11K		1	32
Part 2: Image-centric Data					
CIRR	TI→T	16K		1	32
Fashion200K	T→I	4K		1	32
Nights	I→I	13K		1	32
OVEN	TI→TI	20K	30K	1	64
OVEN	TI→T	20K	30K	1	64
VisualNews	T→I	40K	60K	1	64
EDIS	T→TI	12K		1	32
FashionIQ	TI→I	4K		1	32
MSCOCO	I→T	4K		1	32
REMUQ	TI→T	5K		1	32
WebQA	T→TI	12K		1	32
LLAVA	TI→T	20K	30K	1	64
EVQA	TI→TI	20K	30K	1	64
CC3M	T→I	200K	300K	3	32
CC3M	I→T	100K	200K	2	64
Laion	T→I	300K	500K	3	32
Laion	I→T	200K		2	64
ImageNet	I→T	100K	200K	2	32
VL3-Syn7M (short)	T→I	300K	500K	3	32
VL3-Syn7M (short)	I→T	100K	200K	2	64
VL3-Syn7M (detailed)	T→I	200K	300K	3	32
VL3-Syn7M (detailed)	I→T	100K	200K	2	64
VISTA	TI→I	100K		2	32
VISTA	T→TI	20K		1	64
Part 3: Video-centric Data					
VAST	T→V	1.6M		0	32
InternVid-FLT	T→V	1.7M		0	32
PE-Video	T→V	104K		1	64
WebVid	T→V	5.4M		0	32

Table A.4: Configuration of UVRD. Sample sizes are identical for both 3B and 7B models. Notes: (a) Task: T=Text, I=Image, V=Video (Query→Corpus). (b) Neg.: Explicit hard negatives per positive. '0' denotes in-batch negatives only.

Dataset	**Task**	**Size**	**Neg.**	**BS**
UVRD-T2T	T→T	100K	0	64
UVRD-T2I	T→I	210K	0	32
UVRD-T2V	T→V	879K	1	64
UVRD-TI2V	TI→V	89K	1	64
UVRD-TV2V	TV→V	35K	1	64
UVRD-I2V	I→V	200K	0	64
UVRD-V2V	V→V	36K	0	64

Table A.5: Total number of instances for our 3B and 7B models across all data categories.

Data Category	**Total Sample Size**	
	3B Model	**7B Model**
Collected: Text-only	380K	580K
Collected: Image-centric	1.82M	2.80M
Collected: Video-centric	8.80M	8.80M
Synthesized: UVRD (All Modalities)	1.55M	1.55M
Total	**12.55M**	**13.73M**

less model and evaluation compatibility.

The expressiveness of this data construction pipeline stems from polymorphic specialization via inheritance, encapsulated in three core strategies:

Modality as Configuration. Modality is a dynamic parameter. By overriding a single method, a dataset effortlessly transitions between modalities (text, image, video, or composite), transforming benchmarks like MSRVTT into image-to-video tasks with minimal code.

Task Derivation via Inheritance. Complex tasks are composed of simpler ones. A text+image-to-video task inherits its base structure and extends the query schema. Variants are derived by overriding specific data-access methods, turning benchmark creation into a modular, hypothesis-driven workflow.

Engineering for Scale and Integrity. Scalability and data quality are first-class principles. Large datasets (e.g., InternVid-FLT) leverage chunked, parallel processing. Crucially, proactive curation is embedded: automated validators check video integrity (resolution, frame count), while repair mechanisms fix common errors using `ffmpeg`, ensuring failures reflect retrieval challenges, i.e., not data corruption.

A.1.5 Prompt for Synthetic Video Retrieval Data Generation

To ensure high-quality, diverse, and controllable synthetic video retrieval data generation at scale, we design and deploy a suite of structured prompts to instruct MLLMs for captioning. Our pipeline operates in four parts:

1. **Raw (Or Weakly Annotated) Video (Re-) Captioning**: Enhance raw or uncaptioned videos with rich, diverse textual descriptions.

2. **Text-Image Composed Retrieval**: Generate queries that combine reference images with video content for fine-grained retrieval.

3. **Text-Video Composed Retrieval**: Generate queries that combine short reference clips with target videos for practical temporal or perspective-based retrieval.

4. **Frame Image Captioning**: Annotate individual video frames with dynamic-aware captions for auxiliary training signals.

All prompts enforce strict output formatting, factual grounding, and stylistic diversity to ensure dataset quality and coverage. `{raw_caption}` represents an optional, potentially low-quality human-provided or auto-generated caption associated with the video. Placeholders (e.g., `{readability}` and `{education_level}`) are dynamically instantiated during batch generation.

Besides, we can generate text-to-text retrieval data from the multiple video/frame captions by randomly matching any two sentences in the caption list.

Synthetic Video Captioning Prompt

```
Generate 5 distinct and high-quality ENGLISH captions for the provided video.
    Please first visually understand the video file and analyze the video frame-
    by-frame in depth before captioning.

The original video caption is {raw_caption}. (Ignore if empty.)

Each caption must be:
  1. The final answer MUST only be a JSON dict of captions where the key is the
    caption number and the value is a single-paragraph caption.
  2. Factually accurate and descriptive - include only what is clearly visible
    or reasonably inferable from the video.
  3. Focus exclusively on visible content; do not mention absences or speculate
    about unseen elements.

Content Requirements (per caption):
  1. Spatial Details (30-60%): Describe location, setting, key environmental
    elements, objects, and their spatial relationships. Include notable visual
    features.
  2. Temporal Flow (30-60%): Capture event sequence and action dynamics -
    movements, interactions, transitions - as they unfold over time.
  3. Theme/Background/Style/Meaning/Highlight/Camera (0-20%): Describe
    observable emotional tone, narrative style, thematic elements, highlighting
    frames, or significant camera movements/angles.
  4. Others (0-10%).
```

```
Key Guidelines:
  1. Do not invent fictional elements, dialogue, or backstory not visible in the
     video.
  2. Use the following varied sentence styles (one per caption):
       - concise and punchy summary,
       - spatial-temporal richly descriptive,
       - abstract understanding,
       - keywords-only,
       - partially relevant information.
  3. Ensure diversity: avoid repetition in wording, focus, or rhythm; at least
     one caption must be <20 words and one >100 words; balance objectivity with
     vivid sensory language; randomly omit minor details in 1 or 2 captions.
  4. Ensure readability matches {readability} and is appropriate for {
     education_level} readers.

Now generate these captions in strict JSON format:
{
   "1": <caption text 1: concise and punchy summary, 10-25 words>,
   "2": <caption text 2: spatial-temporal richly descriptive, 80-200 words>,
   "3": <caption text 3: abstract understanding, 30-100 words>,
   "4": <caption text 4: keywords-only, 5-30 words>,
   "5": <caption text 5: partially relevant information, 10-70 words>
}
```

Synthetic Text-Image Composed Video Retrieval Prompt

```
You are an information retrieval expert specializing in high-value text-image
    composed queries. Your sole objective is to generate queries that
    significantly enhance video retrieval performance by effectively combining a
     reference image and video content.

## What Makes a HIGH-QUALITY Query (Non-Negotiable)
A truly valuable query must satisfy ALL of these criteria:

1. COMBINATION NECESSITY (Most Critical)
   - The query MUST become meaningless or significantly less specific if either
    the image or video is removed.
   - Example (HIGH-QUALITY): "the person FROM REFERENCE IMAGE wearing red jacket
     now skiing"
   - Example (LOW-QUALITY): "a person skiing" (works without image)

2. SEMANTIC PRECISION
   - Must accurately reflect BOTH the visual content of the reference image AND
    the video.
   - Must reference at least ONE specific visual attribute from the image (not
    generic descriptions).
   - Example (HIGH-QUALITY): "matching the blue hat FROM PHOTO, now running
    through park"
   - Example (LOW-QUALITY): "someone similar to image moving" (too vague)

3. RETRIEVAL EFFECTIVENESS
   - Must narrow search results by at least 50\% compared to text-only queries.
   - Must contain actionable constraints that differentiate from 90\% of videos
    in the database.
   - Example (HIGH-QUALITY): "the woman WITH PONYTAIL FROM IMAGE entering
    building at 2PM"
   - Example (LOW-QUALITY): "a woman walking" (too broad)
```

```
4. LOGICAL COHERENCE
   - Must maintain subject-verb-object consistency across modalities.
   - Must avoid semantic contradictions between image attributes and video
    actions.
   - Example (HIGH-QUALITY): "the dog FROM REFERENCE PHOTO chasing a ball"
   - Example (LOW-QUALITY): "the red jacket FROM IMAGE running down hill" (
    jackets don't run)

5. PRACTICAL UTILITY
   - Must solve a real-world ambiguity that neither text nor image could resolve
     alone.
   - Must reflect how actual users would express their information need.
   - Example (HIGH-QUALITY): "same person AS IN PHOTO but wearing blue instead
    of red"
   - Example (LOW-QUALITY): "the image shows a person and the video shows action
    " (no real combination)

## High-Value Query Generation Framework
Follow this structured approach:

1. DEEP ANALYSIS PHASE (Mandatory)
   a) Reference Image Analysis:
      - Identify 1-4 SPECIFIC visual attributes (e.g., "red jacket", "ponytail",
     "blue hat")
      - Determine primary subject with discriminative features
      - Note what CANNOT be determined from image (e.g., action, scene)

   b) Video Content Analysis:
      - Identify primary action using precise verbs (e.g., "running", "entering
    ", "chasing")
      - Note scene context and temporal elements
      - Determine what CHANGES from the static image reference

2. COMBINATION STRATEGY SELECTION
   Choose ONE primary strategy:

   A) IDENTITY PRESERVATION + ACTION CHANGE
      - Structure: [Binding phrase] + [Image attribute] + [Video action]
      - Example: "the man FROM REFERENCE IMAGE in blue shirt now running"

   B) IDENTITY PRESERVATION + SCENE MIGRATION
      - Structure: [Binding phrase] + [Image attribute] + [Scene transition]
      - Example: "same person AS IN PHOTO moving from office to park"

   C) IDENTITY PRESERVATION + NEGATIVE CONSTRAINT
      - Structure: [Binding phrase] + [Negative constraint] + [Video state]
      - Example: "not wearing red jacket FROM IMAGE but blue, walking"

   D) RELATIONAL TRANSFER
      - Structure: [Binding phrase] + [Relationship description]
      - Example: "the dog FROM REFERENCE PHOTO chasing a ball"

3. QUALITY ENHANCEMENT TECHNIQUES
   - Binding Precision: Use "FROM REFERENCE IMAGE", not "like the picture"
   - Attribute Specificity: Use concrete features ("red jacket", not "clothing")
   - Action Verbs: Use present continuous tense ("running", not "runs")
   - Context Enrichment: Add 1 relevant scene descriptor ("in park", "near
    building")
   - Noise Handling: For low-quality inputs, use "resembling" but maintain
```

```
    specificity
   - Audience Adaptation: Ensure readability matches {readability} and suits {
    education_level} readers.

4. MANDATORY QUALITY CHECK
   Before finalizing, verify ALL:
   - [ ] Explicit binding to reference image
   - [ ] References SPECIFIC visual attribute from image
   - [ ] Describes DYNAMIC ELEMENT not in static image
   - [ ] Loses specificity if image removed
   - [ ] Contains actionable retrieval constraint
   - [ ] Maintains logical subject-action consistency
   - [ ] Solves real-world ambiguity

## What to AVOID
- Generic descriptions ignoring image specificity
- Redundant mentions of obvious image content
- Semantic contradictions (e.g., "the jacket is running")
- Overly precise details not visible in inputs
- Standalone video descriptions without image binding
- Vague terms: "something", "thing", "area"
- Excessive length without added value (>50 words)

## Output Format
- Output ONLY a JSON object with one key: "query"
- Value must be the generated query sentence
- No additional text or formatting

Generate the query for:
Reference Image: <image_input>
Video Clip: <video_input>
```

Synthetic Text-Video Composed Video Retrieval Prompt

```
You are an information retrieval expert specializing in practical text-video
    composed queries. Your objective is to generate queries that effectively
    combine a short reference clip with a target video for real-world retrieval.

## Practical Context Understanding
In real-world scenarios:
- Reference is typically a SHORT CLIP (2-10 seconds)
- Clip is usually HIGHLY RELEVANT to target video (same source or similar
    content)
- Common relationships: temporal continuation, perspective variation, quality
    differences, minor action variations
- Query should reflect how users actually search

## What Makes a HIGH-QUALITY Query (Practical Focus)
Must satisfy these criteria:

1. USEFUL COMBINATION (Most Important)
   - Leverages reference clip to specify what text alone cannot
   - Example (HIGH-QUALITY): "the same person FROM REFERENCE CLIP continuing to
    run after the jump"
   - Example (LOW-QUALITY): "a person running" (ignores reference)

2. PRACTICAL PRECISION
   - References at least ONE observable feature from reference clip
```

```
   - Example (HIGH-QUALITY): "matching the red jacket FROM REFERENCE, now
    entering building"
   - Example (LOW-QUALITY): "someone similar to clip moving" (too vague)

3. REAL-WORLD UTILITY
   - Helps find videos difficult to retrieve with text alone
   - Example (HIGH-QUALITY): "same action AS IN REFERENCE but from front angle"
   - Example (LOW-QUALITY): "the clip shows action" (no retrieval value)

4. NATURAL EXPRESSION
   - Sounds like how a real user would phrase it
   - Example (HIGH-QUALITY): "what happens right after this moment?"
   - Example (LOW-QUALITY): "temporal continuation of the current visual
    sequence" (too academic)

## Practical Query Generation Framework

1. REFERENCE CLIP ANALYSIS
   - Identify 1-3 KEY OBSERVABLE FEATURES (e.g., "red jacket", "starting pose",
    "mid-action")
   - Determine most distinctive visual or action element
   - Note what is CLEARLY VISIBLE (avoid guessing)

2. TARGET VIDEO RELATIONSHIP ASSESSMENT
   Determine relationship type:

   A) TEMPORAL CONTINUATION
      - Reference is earlier part of same sequence
      - Query focus: "what happens next" or "continuing action"

   B) PERSPECTIVE VARIATION
      - Same action from different angle/view
      - Query focus: "same action from different angle"

   C) QUALITY/CONDITION VARIATION
      - Same action with different lighting/resolution
      - Query focus: "same scene in better lighting"

   D) MINOR ACTION VARIATION
      - Slightly different execution of similar action
      - Query focus: "same person but running instead of walking"

3. QUERY CONSTRUCTION
   - Start with binding phrase: "FROM REFERENCE CLIP", "AS IN REFERENCE", etc.
   - Reference 1-2 specific observable features from clip
   - Describe relationship to target video clearly
   - Keep natural and practical (5-15 words typically)
   - For temporal: focus on "what happens next"
   - For perspective: specify desired viewpoint
   - For quality: specify desired condition
   - For action: specify the change

4. PRACTICAL QUALITY CHECK
   Ask before finalizing:
   - Would this help me find what I'm looking for?
   - Does it add value beyond describing the target?
   - Would a real user phrase it this way?
   - Is everything mentioned clearly visible in reference?

## What to AVOID
```

```
- Overly academic or technical language
- References to features NOT clearly visible
- Excessive precision about timing ("exactly 3.2 seconds later")
- Queries that work equally well without reference
- Generic descriptions: "similar video", "related content"
- Making unsupported assumptions

## Output Format
- Output ONLY a JSON object with key: "query"
- Value must be the generated query string
- No additional text, explanations, or formatting

Generate the query for:
Reference Clip: <reference_clip_input>
Target Video: <target_video_input>
```

Synthetic Frame Captioning Prompt

```
Generate 5 distinct and high-quality ENGLISH captions for the provided image (a
    frame extracted from a video) based solely on visual content. Please first
    visually understand the image and analyze it in depth before captioning.

Each caption must be:
  1. The final answer MUST only be a JSON dict of captions where the key is the
    caption number and the value is a single-paragraph caption.
  2. Factually accurate and descriptive - include only what is clearly visible;
    captions should also be consistent with the short-term video context.
  3. Focus exclusively on visible content; do not mention absences or speculate
    about unseen context.

Content Requirements (per caption):
  1. Spatial Details (50-70%): Describe location, setting, key objects, spatial
    relationships, colors, lighting, composition. Include instantaneous action
    states derived from visible posture/motion cues.
  2. Temporal Snapshots (0-30%): Describe the frozen moment's temporal state (
    movements, interactions, transitions) ONLY if visually provable. Avoid
    implying sequence, duration, or speed.
  3. Theme/Style/Composition (10-20%): Cover emotional tone, camera angle,
    lighting style, or artistic elements directly observable.
  4. Others (0-10%).

Key Guidelines:
  1. Do not invent fictional elements or backstory. If action is ambiguous,
    describe neutrally. Never use future/past tense or speculative phrases.
  2. Use varied sentence styles (randomly assign one per caption):
       - concise spatial-temporal snapshot,
       - spatially rich descriptive,
       - abstract spatial interpretation,
       - keywords with temporal anchors,
       - minimalist spatial focus.
  3. Ensure diversity: avoid repetition in wording, focus, or rhythm; at least
    one caption <20 words and one >100 words; balance objectivity with vivid
    sensory language; randomly omit minor details in 1-2 captions.
  4. Ensure readability matches {readability} and is appropriate for {
    education_level} readers.

Now generate these captions in strict JSON format:
{
```

```
    "1": <caption text 1: concise and punchy summary, 10-25 words>,
    "2": <caption text 2: spatial-temporal richly descriptive, 50-200 words>,
    "3": <caption text 3: abstract understanding, 30-100 words>,
    "4": <caption text 4: keywords-only, 5-30 words>,
    "5": <caption text 5: partially relevant information, 10-70 words>
}
```

Prompt Utilization. These prompts are designed for a scalable, schema-constrained generation pipeline for synthesizing diverse video-centric annotations and queries. The system unifies four tasks under a single modular framework. Each task is governed by a structured prompt template with dynamic control slots (e.g., readability, education level), ensuring linguistic and semantic diversity. Input modalities (image, video, or both) are automatically routed and embedded via a distributed multimodal LLM (Keye-VL-8B [19], 32K context), with outputs rigorously validated against JSON schemas for structural correctness. The pipeline supports sharded, resumable batch generation with quality-aware sampling, enabling the production of millions of grounded, stylistically varied synthetic instances.

A.1.6 Model Architecture Details

The GVE model is architecturally derived from Qwen2.5-VL [20], repurposed as a fixed-length multimodal encoder by removing its autoregressive head. Its core function is to map arbitrarily composed inputs—text, image, or video—into a shared d-dimensional embedding space, preserving cross-modal alignment inherited from pretraining.

Input fusion begins with tokenization and visual encoding. Text is converted into a sequence of token IDs $\mathbf{X}_t \in \mathbb{Z}^{B\times T_t}$, while images and video frames are encoded into visual token sequences $\mathbf{X}_v \in \mathbb{R}^{B\times T_v\times d}$, where $T_v = \text{THW}/p_s^2$ for images, and $T_v = K\cdot(\text{THW}/p_s^2)$ for videos with K uniformly sampled frames. Note that frame embeddings will be added with absolute time encoding. Critically, K must satisfy $K \bmod p_t = 0$ (default: $p_t = p_s = 2$) to maintain alignment with the vision encoder's 3D spatiotemporal grid. The processor then injects these visual tokens into the textual sequence by replacing placeholder tokens

(<image>, <video>), producing a fused input $\mathbf{X}_{\text{fused}} \in \mathbb{R}^{B\times T\times d}$, where $T = T_t + \sum T_v$. This scatter-based fusion preserves positional coherence and enables interleaved modality composition, which is essential for compositional query understanding in multimodal embeddings. The final embedding $\mathbf{e}^{(i)} \in \mathbb{R}^d$ for the i-th instance is extracted from the last attended token in the sequence:

$$\mathbf{e}^{(i)} = \frac{\mathbf{h}^{(i)}_{p_i}}{\|\mathbf{h}^{(i)}_{p_i}\|_2}, \quad \text{where } p_i = \max\{j \mid \mathbf{M}^{(i)}_j = 1\},$$

and $\mathbf{M}^{(i)}$ is the attention mask for instance i. This position corresponds to the EOS token in left-padded sequences or the final non-pad token in right-padded ones. The choice is motivated by the observation that in instruction-tuned MLLMs, the final token often encapsulates the model's response intent, making it semantically aligned with the user's retrieval goal.

A.1.7 Training Implementation and Hyperparameters

Parameter-Efficient Tuning. To facilitate parameter-efficient fine-tuning (PEFT), we employ Low-Rank Adaptation (LoRA) [21]. Our strategy involves a targeted application of LoRA to the language components of the model, specifically the `q_proj`, `v_proj`, `k_proj`, `up_proj`, `down_proj`, and `gate_proj` modules. Crucially, the entire visual backbone and the base token embedding layer are kept frozen. This approach focuses adaptation on high-level semantic and cross-modal reasoning while preserving the powerful pretrained visual features. We enable FlashAttention-2 [22] to accelerate training and reduce memory. The final embedding for each input is obtained via last-token pooling on the last hidden layer, followed by L2 normalization to project embeddings onto the unit hypersphere for stable cosine similarity computations.

Optimizer and Training Dynamics. The model is trained using the AdamW [23] optimizer with a learning rate of 3×10^{-5} and a weight decay of 0.1. A cosine learning rate scheduler is used. Training is performed in BFloat16 (`bf16`) mixed-precision. The entire training process is managed under the DeepSpeed framework. For video inputs, we uniformly sample 8 frames per clip at a rate of 1.0 FPS. By default, we use 32 NVIDIA A100 GPUs, each with 80GB of memory. Therefore, the overall batch size is at least 1024.

Memory Optimization and Distributed Strategy. To manage GPU memory, we enable gradient checkpointing, which recomputes intermediate activations during the backward pass. While our framework supports more advanced techniques like Gradient Cache, it was disabled in favor of this standard approach. To leverage our multi-GPU setup, we enable cross-device negative sharing. This strategy gathers embeddings from all GPUs, effectively multiplying the pool of in-batch negatives by the number of devices. This enriches the negative set for the contrastive loss computation on each GPU, leading to a stronger training signal without increasing per-device memory load.

Contrastive Learning and Stability. The core of our training is an InfoNCE-style contrastive loss with a temperature of 0.03. The use of cross-device negatives starts from the first step. We also enhance our training logs with contrastive-specific metrics, including the average scores of positive pairs and the average margin between positive and hard-negative pairs, providing crucial insights into the model's learning dynamics.

Hyperparameter Summary. A comprehensive summary of all key hyperparameters is provided in Table A.6.

Table A.6: Key hyperparameters used for model training.

Parameter	Value
Model & LoRA Architecture	
Base Model	`Qwen2.5-VL-3B-Instruct` or `Qwen2.5-VL-7B-Instruct`
PEFT Method	LoRA
LoRA Rank (r)	16
LoRA Alpha (α)	32
LoRA Dropout	0.1
LoRA Target Modules	`q_proj, v_proj, k_proj, up_proj, down_proj, gate_proj`
Frozen Components	Visual Backbone, Token Embeddings
Optimizer & Training	
Optimizer	AdamW
Learning Rate	3×10^{-5}
Weight Decay	0.1
LR Scheduler	Cosine
Training Epochs	3
Precision	BF16
Gradient Checkpointing	Enabled
Seed	42
Contrastive Learning	
Temperature	0.03
In-batch Negatives	Enabled
Cross-Device Negatives	Enabled
Data & Preprocessing	
Video Frames per Clip	8
Video Sampling Rate	1.0 FPS
Dataloader Workers	1

A.1.8 Baseline Details

Here we present more details of baseline models tested on our benchmark in Table A.7, including full model names, abbreviations, architectures, sizes, and training data types. Based on these properties, we analyze to discover potential performance knowledge and dependencies in our experimental part.

Table A.7: Model Abbreviations, Architectures, Parameter Sizes, and Training Data Types. The checkmark (✓) indicates the model was trained on the corresponding data pair types.

Full Model Name	Abbreviation	Architecture	Size
CLIP4Clip	CLIP4Clip	CLIP-based	87M
ViCLIP	ViCLIP	CLIP-based	0.4B
VideoCLIP-XL	VideoCLIP-XL	CLIP-based	0.4B
LanguageBind-Video-Huge-V1.5	LanguageBind	CLIP-based	1.2B
InternVideo2-Stage2-1B	InternVideo2-1B	CLIP-based	1.4B
InternVideo2-Stage2-6B	InternVideo2-6B	CLIP-based	6.4B
gme-Qwen2-VL-2B-Instruct	GME-2B	MLLM-based	2.2B
Unite-Base-Qwen2-VL-2B	Unite-2B	MLLM-based	2.2B
VLM2Vec-V2.0	VLM2Vec-V2	MLLM-based	2.2B
BGE-VL-v1.5-mmeb	BGE-VL	MLLM-based	7.6B
UniME-LLaVA-OneVision-7B	UniME-7B	MLLM-based	8.0B
B3-Qwen2-7B	B3-7B	MLLM-based	8.3B
gme-Qwen2-VL-7B-Instruct	GME-7B	MLLM-based	8.3B
Unite-Base-Qwen2-VL-7B	Unite-7B	MLLM-based	8.3B
GVE-3B	GVE-3B	MLLM-based	3.8B
GVE-7B	GVE-7B	MLLM-based	8.3B

Abbreviation	Text-Text	Text-Image	Text-Video
CLIP4Clip	-	-	✓
ViCLIP	-	-	✓
VideoCLIP-XL	-	-	✓
LanguageBind	-	-	✓
InternVideo2-1B	-	-	✓
InternVideo2-6B	-	-	✓
GME-2B	✓	✓	-
Unite-2B	✓	✓	✓
VLM2Vec-V2	✓	✓	✓
BGE-VL	✓	✓	-
UniME-7B	✓	✓	-
B3-7B	✓	✓	-
GME-7B	✓	✓	-
Unite-7B	✓	✓	✓
GVE-3B	✓	✓	✓
GVE-7B	✓	✓	✓

A.1.9 Training Dynamics

We monitor four key metrics during training: *Training Loss*, *Mean Score*, *Max Negative Gap*, and *Mean Positive Score* in Figure A.2 and Figure A.3. To ensure robust visualization, we mitigate outlier effects, followed by 200-step moving average smoothing. Original trajectories (subsampled every 100 steps) are plotted alongside smoothed trends, with ±1 standard deviation bands indicating local volatility.

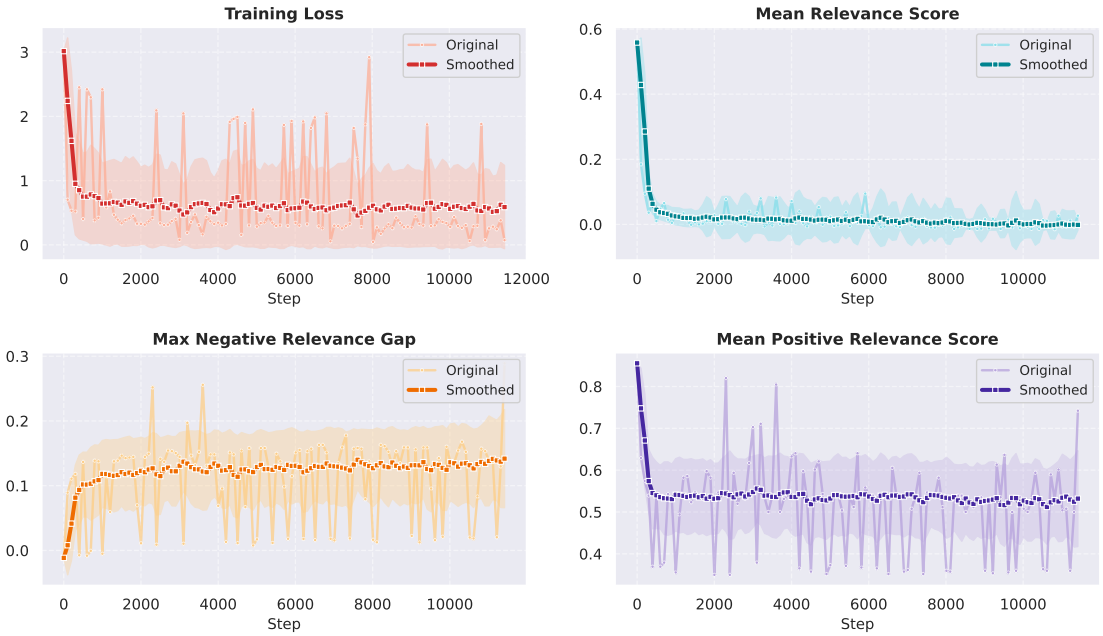


Figure A.2: Training dynamics of `GVE-3B` across four metrics.

A.1.10 Experiments of Training-time Scaling: More Results of Data Scaling

Along with Section 3.4.2, we depict more experiments for six abilities in data scaling for `GVE-3B` and `GVE-7B` in Figure A.4. Shaded bands show ±1 std; dashed curves are log-linear fits for visual guidance. X-axis is log-scaled to reflect scaling law dynamics.

We fit a logarithmic scaling law $y = a \ln x + b$ to quantify gains per decade of data

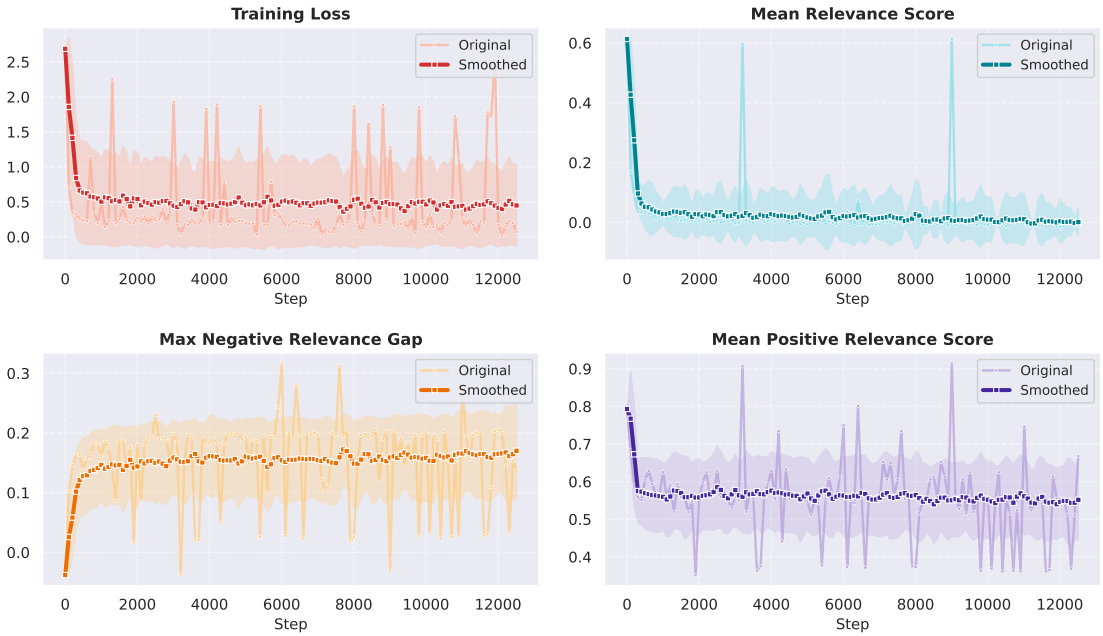


Figure A.3: Training dynamics of GVE-7B across four metrics.

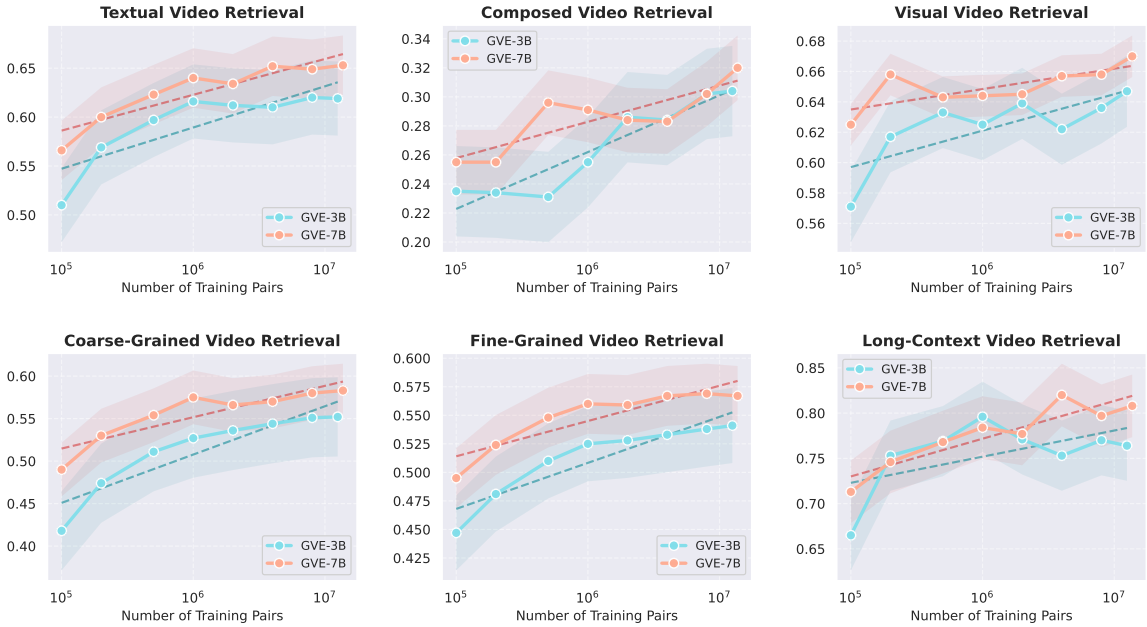


Figure A.4: Performance effect from data scaling for GVE-3B and GVE-7B in detail.

(10× increase). GVE-3B consistently shows higher relative gains: in compositional retrieval (CMP), it improves by $+0.039$ (14.7%), nearly double GVE-7B's $+0.025$ (8.7%); in coarse-grained tasks (CG), it gains $+0.057$ (11.1%), far exceeding GVE-7B's $+0.037$ (6.6%). Textual (TXT) and fine-grained (FG) retrieval follow similar trends, with GVE-3B improving by $+0.042$ (7.1%) and $+0.040$ (7.9%), respectively, versus GVE-7B's $+0.037$ (5.8%) and $+0.031$ (5.6%). Visual retrieval (VIS) scales weakly for both ($+0.024/3.8\%$ for GVE-3B, $+0.013/2.1\%$ for GVE-7B). Notably, only in long-context retrieval (LC) does GVE-7B outperform GVE-3B in both absolute ($+0.042$ vs. $+0.029$) and relative gain (5.4% vs. 3.8%).

This reveals a task-dependent scaling trade-off: smaller models (GVE-3B) scale more efficiently in semantic and compositional tasks, while larger models (GVE-7B) uniquely excel in long-context modeling — suggesting that model size should be chosen not only for capacity, but for alignment with the data-scaling profile of the target task.

A.1.11 Experiments of Test-time Scaling

We investigate the impact of scaling two test-time parameters: the number of sampled frames and the maximum tokens per frame (i.e., effective resolution). We analyze each in isolation: when scaling the frame count from 8 to 48, the maximum token count is fixed at 200; conversely, when scaling the token count from 200 to 800, the frame count is fixed at 8.

Temporal Scaling: the Number of Sampled Frames. As shown in Figure A.5, increasing the number of sampled frames generally improves performance. The most substantial gains appear in the Long-Context (LC) task (GVE-3B: +19.6%, GVE-7B: +12.8%), underscoring the value of denser temporal sampling for long-range reasoning. However, the Compositional (CMP) task shows a slight performance degradation, suggesting its sensitiv-

ity to potentially redundant visual cues from excessive frames. Furthermore, while GVE-7B consistently outperforms GVE-3B, the performance gap narrows as frames increase (from 0.029 to 0.024 on average scores across datasets), indicating that the larger model is more efficient at extracting information from sparser inputs.

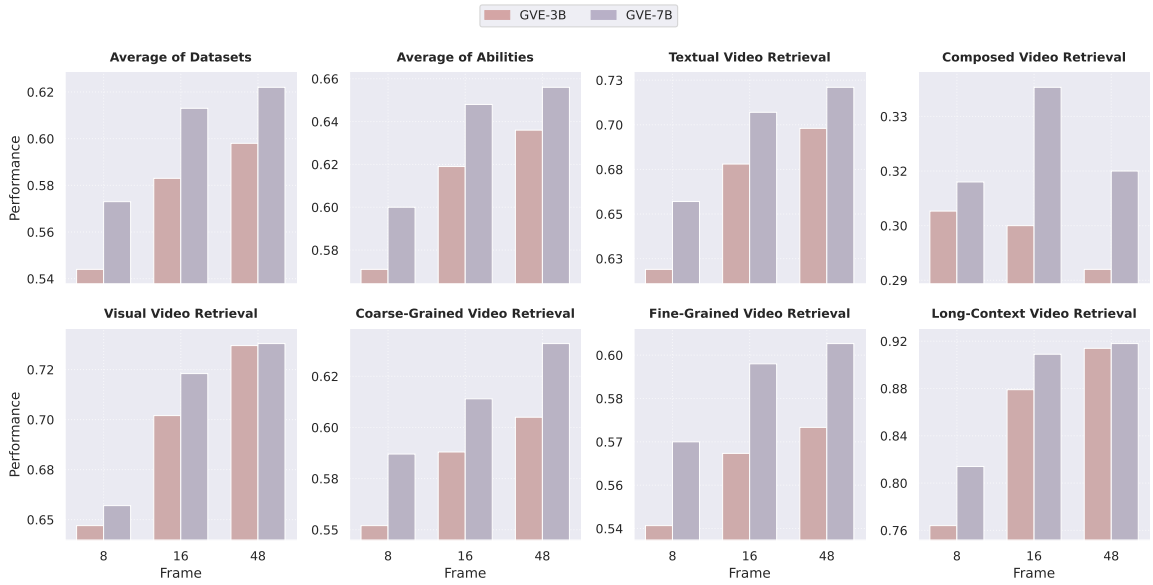


Figure A.5: Model performance as a function of the number of sampled frames at inference time (max tokens fixed at 200).

Spatial Scaling: the Maximum Tokens Per Frame. In contrast, increasing the token budget yields non-monotonic returns (Figure A.6). Performance for most tasks peaks around 400 tokens and then declines. This suggests that for semantic retrieval tasks, there is an optimal point of spatial detail. Scaling beyond this introduces a massive influx of fine-grained visual features that are largely redundant for the task. This over-saturation of information can dilute the effective semantic signal within the LLM's context window, making it more challenging for the model to distill the core visual concepts into the final embedding. The CMP task's performance again deteriorates with more tokens, reinforcing its sensitivity to input redundancy.



Figure A.6: Model performance versus the maximum number of tokens per frame (frame count fixed at 8), determining the resolution of frames in resizing.

Key Findings and Implications. Our scaling analysis reveals four key findings:

- **Temporal Scaling (More Frames):** Provides a reliable, though diminishing, performance boost, especially for long-range tasks.
- **Spatial Scaling (More Tokens):** Exhibits a clear optimal point (approx. 400 tokens). Exceeding it degrades performance, indicating a trade-off between detail and distraction.
- **Model Scale Efficiency:** The larger model (GVE-7B) benefits less from input scaling, suggesting a greater intrinsic capacity to process sparse inputs effectively.

For test-time enhancement, temporal scaling is a robust strategy, whereas naive spatial scaling is not. This distinction underscores a critical insight: performance is not merely a function of total input data, but of its effective composition. It strongly motivates a shift from brute-force data scaling towards adaptive input mechanisms, such as dynamic token or frame selection, that can intelligently manage information density.

A.1.12 Human-in-the-Loop Quality Validation of UVRD

To provide a ground-truth assessment of our synthesized data, we conducted a human expert evaluation on a random sample of 100 instances generated from UVRD. As referenced in Section 3.3.1, the validation yielded a 95% **factual accuracy rate**. The 5 erroneous instances were categorized as follows, confirming they were minor and non-critical in nature: 2 instances with minor over/under-claims of visual details (e.g., recognize an object as a laptop/butterfly but it is not obvious); 2 instances with minor inaccuracies in camera motion description (e.g., movement hallucination); 1 instance with a subtle temporal sequencing error (e.g., reversed temporal description). This high fidelity rate validates the effectiveness of our multi-stage quality assurance process in producing reliable training data at scale.

A.1.13 Experimental Analysis of Modality Pyramid

This subsection provides detailed experimental data to quantitatively validate the design choices and overall effectiveness of the Modality Pyramid, as discussed in Section 3.3.2 and Section 3.4.2.

Empirical Validation of Task Hierarchy The bottom-up design of our curriculum is predicated on an assumed hierarchy of task complexity. To empirically ground this assumption, we measured the initial alignment scores of different task types before training. We computed the average cosine similarity for 100 positive pairs per task using the GME-7B [15]. The results, presented in Table A.8, reveal a clear monotonic decrease in similarity as task complexity increases, from 0.68 for text-to-text (T2T) down to 0.43 for text-video-composed (TV2V) retrieval. This trend provides strong quantitative evidence for the assumed task hierarchy, justifying the curriculum's prioritization of foundational tasks.

Table A.8: Average cosine similarity of positive pairs across task types at initialization. The scores reveal a natural task difficulty hierarchy, providing an empirical basis for our curriculum design. T: text, I: image, V: video.

Task Type	T2T	T2I	T2TI	I2TI	TI2TI	T2V	TI2V	TV2V
Similarity	0.68	0.65	0.59	0.53	0.52	0.46	0.45	0.43

Impact of Temperature Scheduling The curriculum's dynamic task scheduling is governed by a temperature parameter $\sigma(t)$ that increases linearly over time. To analyze its effectiveness, we compared this linear schedule against a fixed, manually tuned constant temperature ($\sigma = 0.5$) for GVE-3B. As shown in Table A.9, the linear schedule achieves slightly superior final performance across datasets. This result suggests that an automated, adaptive scheduling mechanism can match or exceed a carefully tuned static hyperparameter, enhancing the robustness of the training process.

Table A.9: Performance comparison of temperature scheduling strategies at 3B scale.

Scheduling Strategy	Score
Linear Schedule (Ours)	**0.571**
Constant Temperature ($\sigma = 0.5$)	0.569

Table A.10: Video classification accuracy comparison across different models and datasets. For each column: highest score is **bolded**, second-highest is underlined.

Model	**AVG**	**K700**	**UCF101**	**HMDB51**	**SSV2**	**Breakfast**
CLIP4Clip	0.378	0.395	0.596	0.413	0.308	0.176
InternVideo2-6B	**0.526**	**0.554**	0.572	0.480	**0.569**	**0.453**
VLM2Vec-V2	0.393	0.380	0.600	0.409	0.428	0.148
GME-7B	0.374	0.397	0.547	0.479	0.306	0.143
UniME-7B	0.306	0.388	0.377	0.407	0.190	0.166
Unite-7B	0.519	0.537	0.752	**0.534**	0.513	0.261
GVE-3B	0.476	0.489	0.661	0.483	0.471	0.277
GVE-7B	**0.526**	0.540	**0.757**	0.525	0.521	0.289

The effectiveness of this linear schedule lies in its data-driven nature. The prober model implicitly determines task difficulty based on alignment scores, leading to an adaptive

schedule without requiring manual prior assumptions about which tasks are inherently "harder." We believe this data-driven approach is more robust for a universal framework than hand-crafted, task-specific schedules.

A.1.14 Experiments of Video Classification

We test the video embeddings for video classification by following the MMEB-V2 [12]. Note that we do not train on these datasets for zero-shot evaluations, while other baselines may include them in optimization. Specifically, we evaluate video embedding models on five diverse classification benchmarks, including **Kinetics-700 (K700)** [24], **UCF101** [25], **HMDB51** [26], **SomethingSomething-V2 (SSV2)** [27], and **Breakfast** [28], covering fine-grained interactions, open-domain actions, and procedural activities. Each task is cast as video-to-text retrieval over the full label set, using standardized validation/test splits. The results are provided in Table A.10. Our results show that `GVE-7B` and `InternVideo2-6B` achieve the highest mean accuracy (0.526), demonstrating the efficacy of unified multi-modal representations. However, performance remains suboptimal on temporally complex datasets such as Breakfast (best: 0.453) and fine-grained SSV2 (best: 0.569), revealing a critical bottleneck in modeling dynamic physical interactions.

APPENDIX B

SUPPLEMENTARY MATERIAL FOR DEMYSTIFYING SPATIOTEMPORAL SCALING LAWS OF MULTIMODAL VIDEO EMBEDDINGS

This document provides additional results and materials referenced in the main chapter. Table B.1 summarises the correspondence between supplementary sections and main-chapter sections.

Table B.1: Section correspondence between the supplementary and the main chapter.

Supp. Section	Content	Main Paper
§4.2.2	Error analysis & case studies	§4.2
§B.2	Baseline results at 32 frames	§4.3.2
§B.3	Additional scaling analyses	§4.3.3–§4.3.4
§B.4	LLM prompts used in the pipeline	§4.2.2

B.1 Error Analysis & Case Studies

Section 4.3 of the main chapter describes three key failure modes. The corresponding visual case studies are provided below.

(1) Degradation on Long Continuous Temporal Segments. As shown in Figure B.1, the model's performance degrades on long, continuous events. This is because our model, limited by a fixed-length context window (e.g., 16 frames), processes videos in short, independent clips during inference. This segmentation prevents it from capturing the full temporal scope of events that span across multiple clips.

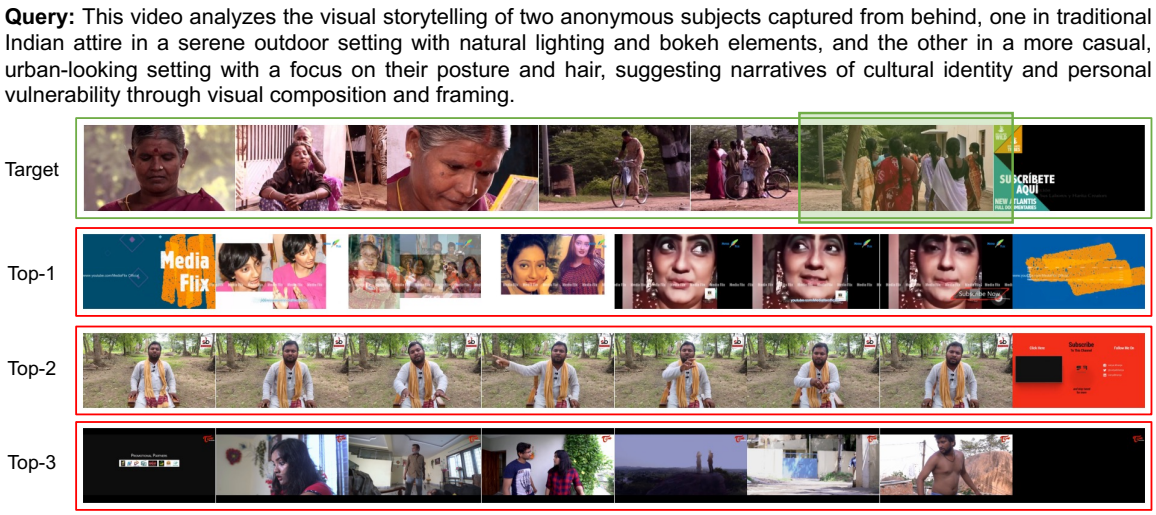


Figure B.1: Example of Error Type 1: *Degradation on Long Continuous Temporal Segments*.

(2) Failure in Fine-Grained Temporal Retrieval. As shown in Figure B.2, this failure occurs when the target event is a brief scene in the background, reflecting difficulties similar to "needle-in-a-haystack" scenarios in video retrieval.

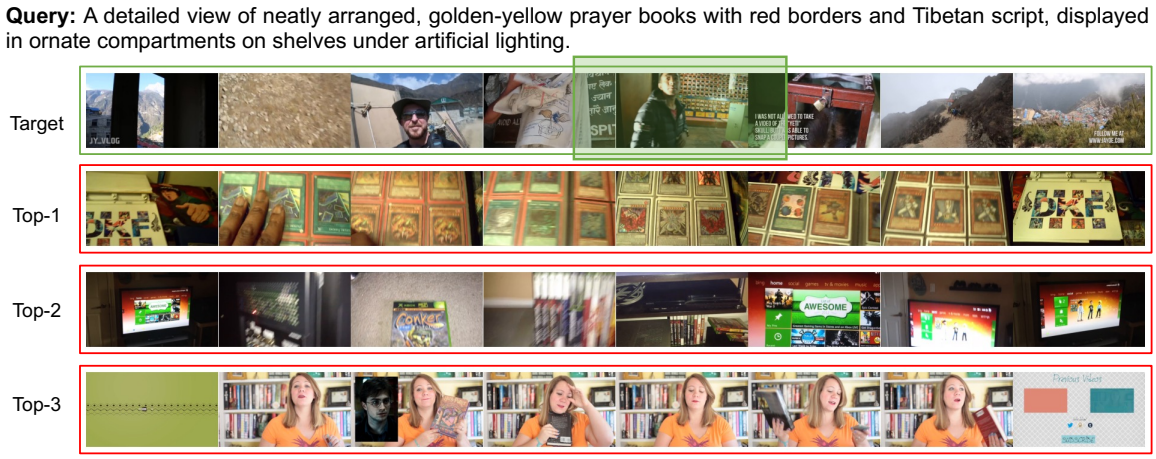


Figure B.2: Example of Error Type 2: *Failure in Fine-Grained Temporal Retrieval*.

(3) Incomplete Satisfaction of Semantic Constraints. As shown in Figure B.3, the model sometimes fails to meet all query conditions. For instance, the top-1 retrieved video satisfies the "speaker" requirement but fails to match the "blue background" constraint.

Similarly, the top-3 retrieved video satisfies most constraints but misses the "speaker in a patterned shirt" condition.

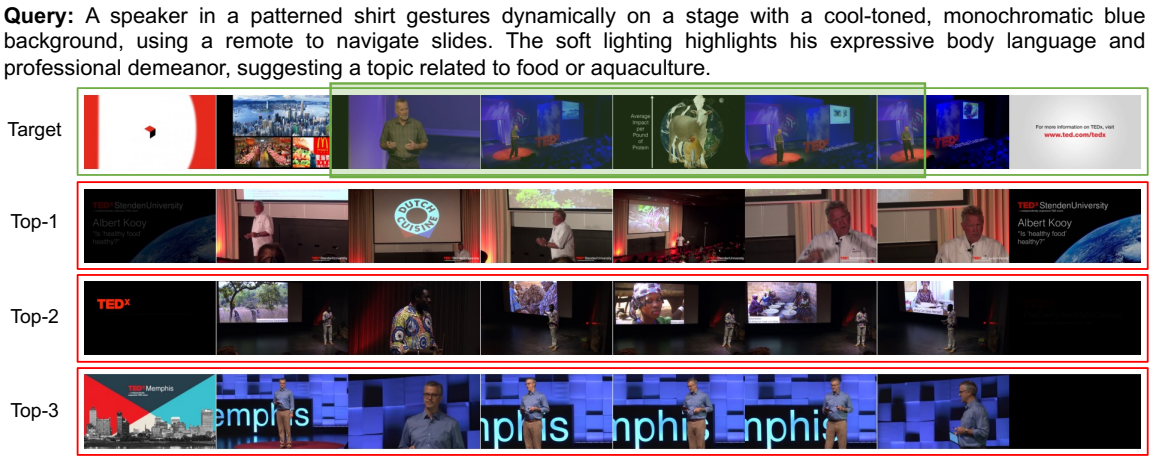


Figure B.3: Example of Error Type 3: *Incomplete Satisfaction of Semantic Constraints*.

B.2 Baseline Results at 32 Frames

The main chapter (§4.3.2) notes that "results at 32 frames are provided in the supplementary material." Table B.2 fills this gap, complementing Table 4.2 (8 F) and Table 4.3 (128 F) of the main chapter.

B.3 Additional Scaling Analyses

B.3.1 Test-Time Spatial Scaling Across Temporal Densities

The caption of Figure 4.3 in the main chapter (§4.3.3) notes that "sensitivity results under varying temporal densities are provided in the supplementary material." Figure B.4 extends Figure 4.3 by sweeping over $F \in \{8, 32, 128\}$ frames per video, whereas Figure 4.3 fixes $F = 32$.

Table B.2: Multi-granularity retrieval performance of all baselines at **32 uniformly sampled frames** per video. **Bold** and underline indicate the best and second-best results within each model group, respectively.

Model	Overall		FN		MD		CR	
	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
InternVideo2-1B	0.1948	0.2653	0.0463	0.0890	0.2641	0.3611	0.5610	0.6512
InternVideo2-6B	0.2815	0.3674	0.0963	0.1620	0.3940	0.5022	0.6429	0.7305
CLIP4Clip	0.3087	0.3955	0.1220	0.1924	0.4302	0.5360	0.6429	0.7286
LanguageBind	**0.4622**	**0.5445**	**0.2284**	**0.3153**	**0.6374**	**0.7211**	**0.7961**	**0.8538**
GME-2B	0.3407	0.4277	0.1445	0.2183	0.4691	0.5741	0.6892	0.7653
Unite-2B	0.3672	0.4528	0.1598	0.2352	0.5123	0.6129	0.7008	0.7744
Ops-MM-2B	0.3832	0.4662	0.1664	0.2412	0.5355	0.6312	0.7302	0.8007
VLM2Vec-V2	0.4002	0.4814	0.1721	0.2446	0.5623	0.6574	0.7578	0.8247
Qwen3-VL-Emb-2B	0.4640	0.5464	0.2091	0.2944	**0.6562**	**0.7417**	0.8237	0.8816
GVE-3B	**0.4679**	**0.5487**	**0.2187**	**0.3062**	0.6475	0.7302	**0.8495**	**0.8949**
GME-7B	0.3835	0.4722	0.1560	0.2390	0.5439	0.6435	0.7453	0.8181
Ops-MM-7B	0.4369	0.5200	0.2085	0.2925	0.5985	0.6879	0.7979	0.8541
UniME-V2	0.4381	0.5206	0.1876	0.2730	0.6183	0.7054	0.8228	0.8766
Unite-7B	0.4600	0.5434	0.2138	0.3010	0.6371	0.7269	0.8388	0.8830
RzenEmbed	0.4751	0.5546	0.2072	0.2934	0.6729	0.7554	0.8682	0.9089
GVE-7B	0.5028	0.5833	0.2505	0.3415	0.6873	0.7683	0.8789	0.9144
Qwen3-VL-Emb-8B	**0.5290**	**0.6017**	**0.2641**	**0.3497**	**0.7301**	**0.7991**	**0.8976**	**0.9301**

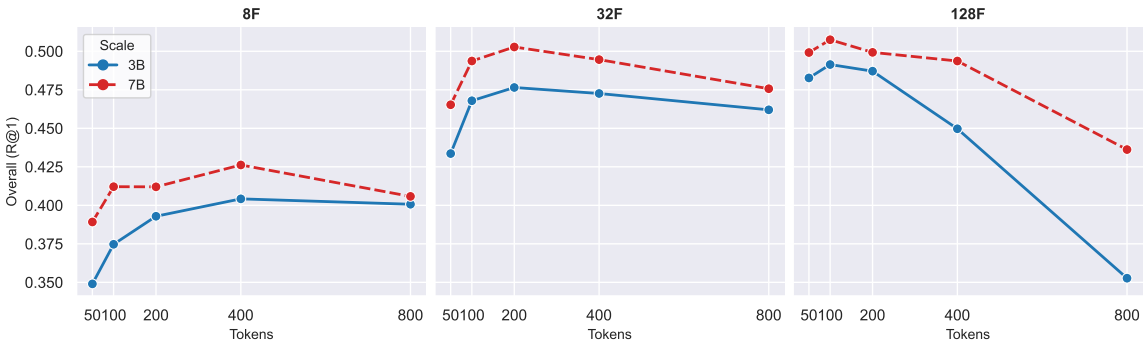


Figure B.4: Test-time spatial scaling sensitivity of GVE at temporal densities $F \in \{8, 32, 128\}$ frames per video. Each panel fixes F and varies the number of spatial tokens T per frame.

B.3.2 Training-Time Spatial Scaling (Evaluated at 8 Test Frames)

Table 3 of the main chapter (§4.3.4) reports training-time spatial scaling evaluated at 4 test frames. Table B.3 provides the complementary results at 8 test frames, with $F_{\text{train}} = 4$ and $T_{\text{train}} \in \{50, 100, 200, 400\}$.

Table B.3: Training-time spatial scaling results with fixed temporal density ($F_{\text{train}} = 4$), evaluated at **8 test frames**. Models trained at varying spatial resolutions ($T_{\text{train}} \in \{50, 100, 200, 400\}$) are evaluated across all spatial test configurations. **Bold** indicates the best result within each group.

Training		Testing		Overall		FN		MD		CR	
Temporal	Spatial	Temporal	Spatial	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
4F	50T	8F	50T	0.2523	0.3349	0.0941	0.1561	0.3438	0.4461	0.5779	0.6746
	100T			0.2567	0.3392	0.1018	0.1620	0.3486	0.4512	0.5672	0.6685
	200T			0.2702	0.3529	0.1130	0.1734	0.3607	0.4647	0.5948	0.6928
	400T			**0.2778**	**0.3610**	**0.1137**	**0.1780**	**0.3759**	**0.4773**	**0.6037**	**0.6991**
4F	50T	8F	100T	0.2773	0.3598	0.1028	0.1673	0.3793	0.4817	0.6322	0.7169
	100T			0.2815	0.3612	0.1065	0.1683	0.3877	0.4851	0.6233	0.7122
	200T			0.2907	0.3709	0.1107	0.1752	0.3993	0.4962	0.6447	0.7292
	400T			**0.3041**	**0.3871**	**0.1220**	**0.1897**	**0.4179**	**0.5169**	**0.6474**	**0.7357**
4F	50T	8F	200T	0.2889	0.3704	0.1037	0.1691	0.4015	0.5008	0.6500	0.7333
	100T			0.2915	0.3721	0.1109	0.1740	0.4000	0.4991	0.6483	0.7341
	200T			0.2970	0.3785	0.1105	0.1767	0.4085	0.5076	0.6670	0.7480
	400T			**0.3062**	**0.3883**	**0.1171**	**0.1851**	**0.4242**	**0.5224**	**0.6634**	**0.7453**
4F	50T	8F	400T	0.2936	0.3738	0.1041	0.1681	0.4090	0.5076	0.6616	0.7423
	100T			0.2977	0.3778	0.1124	0.1754	0.4097	0.5086	0.6607	0.7435
	200T			0.3001	0.3800	0.1171	0.1790	0.4090	0.5085	0.6652	0.7488
	400T			**0.3058**	**0.3861**	**0.1203**	**0.1843**	**0.4198**	**0.5175**	**0.6625**	**0.7469**

B.3.3 Training-Time Temporal Scaling (Evaluated at 100 Spatial Tokens)

Table 4 of the main chapter (§4.3.4) reports training-time temporal scaling evaluated at $T_{\text{test}} = 50$ spatial tokens. Table B.4 provides the complementary results at $T_{\text{test}} = 100$, with $T_{\text{train}} = 50$ and $F_{\text{train}} \in \{4, 8, 16, 32\}$.

Table B.4: Training-time temporal scaling results with fixed spatial density ($T_{\text{train}} = 50$ tokens per frame), evaluated at **100 test spatial tokens**. Models trained at varying frame counts ($F_{\text{train}} \in \{4, 8, 16, 32\}$) are evaluated across six temporal test configurations. **Bold** indicates the best result within each group.

Training		Testing		Overall		FN		MD		CR	
Temporal	Spatial	Temporal	Spatial	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
4F	50T	4F	100T	**0.2426**	**0.3213**	**0.0935**	**0.1498**	**0.3230**	**0.4222**	**0.5708**	**0.6681**
8F				0.2395	0.3189	0.0912	0.1490	0.3213	0.4193	0.5592	0.6609
16F				0.2251	0.3008	0.0861	0.1408	0.2945	0.3891	0.5512	0.6456
32F				0.2286	0.3031	0.0916	0.1440	0.2970	0.3908	0.5503	0.6459
4F	50T	8F	100T	0.3116	0.3984	0.1232	0.1943	0.4336	0.5367	0.6509	0.7438
8F				**0.3188**	**0.4027**	**0.1288**	**0.1979**	**0.4384**	**0.5390**	**0.6741**	**0.7582**
16F				0.2982	0.3828	0.1169	0.1846	0.4116	0.5126	0.6394	0.7345
32F				0.3045	0.3892	0.1258	0.1925	0.4112	0.5151	0.6598	0.7493
4F	50T	16F	100T	0.3549	0.4422	0.1392	0.2160	0.4988	0.6026	**0.7284**	**0.7986**
8F				0.3553	0.4426	0.1400	0.2172	0.5010	0.6035	0.7204	0.7942
16F				**0.3582**	**0.4454**	**0.1405**	**0.2176**	**0.5097**	**0.6114**	0.7124	0.7882
32F				0.3266	0.4116	0.1194	0.1896	0.4611	0.5642	0.6990	0.7793
4F	50T	32F	100T	0.3860	0.4736	0.1619	0.2404	0.5432	0.6462	0.7453	0.8138
8F				0.3805	0.4663	0.1594	0.2362	0.5340	0.6345	0.7409	0.8104
16F				**0.3905**	**0.4783**	**0.1664**	**0.2465**	0.5483	**0.6496**	**0.7480**	**0.8179**
32F				0.3877	0.4741	0.1636	0.2434	**0.5495**	0.6457	0.7302	0.8078
4F	50T	64F	100T	0.4086	0.4923	0.1728	0.2538	**0.5799**	**0.6726**	0.7649	0.8266
8F				0.3960	0.4809	0.1696	0.2485	0.5560	0.6530	0.7551	0.8199
16F				0.4108	0.4961	0.1832	0.2627	0.5739	0.6713	0.7631	0.8284
32F				**0.4148**	**0.4994**	**0.1885**	**0.2693**	0.5739	0.6699	**0.7765**	**0.8347**
4F	50T	128F	100T	0.4130	0.4970	0.1859	0.2660	0.5794	0.6723	0.7507	0.8186
8F				0.4013	0.4853	0.1764	0.2556	0.5618	0.6569	0.7516	0.8143
16F				0.4167	0.5015	0.1840	0.2679	**0.5843**	0.6769	0.7738	0.8330
32F				**0.4226**	**0.5073**	**0.1951**	**0.2782**	**0.5843**	**0.6782**	**0.7801**	**0.8368**

B.4 Prompts Used in the Pipeline

Section 4.2.2 of the main chapter (Step 4: Retrieval Query Synthesis) uses two LLM prompts: one for generating retrieval queries from clip-level captions (§B.4.1), and one for synthesising a consolidated visual description of each video (§B.4.2). Both prompts were executed with Qwen3-32B.

B.4.1 Retrieval Query Synthesis

This prompt is used in Step 4. It receives the aggregated captions of a retrieval unit together with its granularity G and continuity C values, and produces three natural-language queries of increasing descriptive depth.

Retrieval Query Synthesis from Captions

```
## ROLE
You are an expert Search Query Synthesizer. Your primary goal is to read a
    descriptive video caption and generate three high-quality, divergent search
    queries that are strictly grounded in the visual information provided.

## CRITICAL RULES & PROCESSING LOGIC

### 1. Core Mandate: Grounded and Diverse Queries
- **Strictly Visual & Factual**: All generated queries, regardless of their
    complexity, MUST be derived solely from the visual information described in
    the "Raw Caption". Do not invent details not present (e.g., specific brand
    names unless visible, people's names unless specified as visual text).
    Explicitly ignore any mention of audio, speech, voice-overs, sound effects,
    or metadata. Queries must be declarative sentences, not questions or
    commands.
- **Three Distinct Perspectives**: For each valid caption, generate exactly
    three queries. These queries must be genuinely different, reflecting diverse
     user intents and levels of detail.
  - **Query 1 (Librarian/Factual)**: A concise, objective summary of the core
    visual elements. Length: 20-50 words.
  - **Query 2 (Everyday User/Descriptive)**: A more narrative and descriptive
    query, using richer vocabulary to describe the scene, actions, and
    atmosphere. Length: 30-100 words.
  - **Query 3 (Researcher/Analytical)**: A detailed, analytical query that
    examines visual composition, spatial-temporal details, potential technical
    aspects, or the narrative storytelling of the scene. Length: 70-200 words.

### 2. Content Quality Filter: Identify and Discard Invalid Inputs
- Before generating queries, you MUST first assess the "Raw Caption" against the
     following filters. If a caption falls into any of these categories, you
    MUST return an empty list `[]` for "refined_queries".
- **A clip is considered INVALID if its caption describes:**
```

```
  1. **No Visual Content**: An entirely blank, black, white, or single-color
     screen with no other discernible objects, text, or shapes.
  2. **Irretrievably Low Quality**: Content that is explicitly stated to be too
      blurry, corrupted, pixelated, or overexposed to identify any subjects,
     objects, or actions.
  3. **Purely Abstract/Non-Representational**: Content consisting only of
     abstract patterns, fractals, kaleidoscope effects, or color fields without
     any concrete, recognizable subjects. (e.g., "a swirling vortex of
     psychedelic colors").
  4. **Extremely Low Information Density**: A single, simple, static, or
     looping scene with minimal change over a long duration, making the
     generation of three diverse, meaningful queries impossible. (e.g., "a 10-
     minute video of a spinning fan" or "a static image of a single apple on a
     table").
  5. **Dense Text/Code as Primary Content**: Screen recordings that primarily
     show large, scrolling blocks of code, documents, or dense text where the
     main subject is the textual information itself, not a visual narrative or
     action. (This does not apply to simple title cards or logos).
  6. **Refusals or Error Messages**: The caption is an apology, a refusal to
     describe (e.g., "I'm sorry, I cannot describe..."), or an error message.

### 3. Special Handling for Minimalist Content
- If the caption describes a very simple scene (e.g., a plain background) but **
    also contains a key visual element** like a logo, a title card, or a single
    piece of text (e.g., 'LATINO' on a black screen), it is considered **VALID
    **.
- In such cases, your queries must **focus intently on these minimal elements**.
     Describe their visual properties (font, color, style, placement) and, for
    the researcher query, analyze their likely function (e.g., branding, title
    sequence, call-to-action).

## OUTPUT FORMAT
You MUST return ONLY a single, strict JSON object. No explanations or
    conversational text. The required format is:
{"refined_queries": ["Librarian query text", "Everyday user query text", "
    Researcher query text"]}

- If the input is deemed invalid based on the "Content Quality Filter", the
    output MUST be:
{"refined_queries": []}

**Metadata**: Granularity={granularity}, Continuity={continuity}
**Raw Caption**: {caption}

**JSON Output**:
```

B.4.2 Clip Caption Merging

This prompt produces a consolidated visual-only description of each video from its clip-level captions. The output serves as the semantic proxy for building the Clip Relation Graph ($\mathcal{G}$) in Step 2, and is also used during the refinement pass in Step 4 to filter out queries that reference non-visual content.

Clip Caption Merging in Groups

```
## Role & Core Principle
You are a master video summarizer, operating as a **Visual Purist**. Your
    absolute primary goal is to create a single, flowing paragraph that
    describes only what a person would see if they watched the video with the
    sound completely off.

**Your Most Important Rule: Proactive Filtering.**
You MUST critically analyze the input `caption` for each clip and proactively
    filter out any information that is not strictly visual. **Completely ignore
    and do not mention**:
-   **Spoken Content**: Dialogue, narration, interviews, or any transcribed
    speech (e.g., "the narrator explains...", "discusses the importance of...").
-   **Character/Object Names & Backstories**: Proper names of people, places, or
     things (e.g., "Midas the mouse," "the Napa fires"), unless they are clearly
     visible as text in the scene (e.g., a brand name on a product).
-   **Abstract Concepts & Inferred Knowledge**: Intentions, emotions, technical
    jargon (e.g., "'Inverted Single Stroke Four' drum fill"), or specialized
    knowledge that cannot be seen directly (e.g., "the car's engine is powerful
    ").
-   **Audio Cues**: Descriptions of music, sound effects, or tone of voice.

## Input Analysis & Synthesis Logic
Follow this structured three-step process to generate the summary:

**Step 1: Identify the Core Subject & Setting.**
-   Quickly scan all clip captions to determine the recurring central subject (e
    .g., a person, an animal, an object) and the primary environment (e.g., a
    kitchen, a forest, a studio). This forms the foundation of your summary.

**Step 2: Construct the Narrative Flow.**
-   Start your summary by introducing the core subject and setting you
    identified.
-   Chronologically process the clips, describing the **progression of visual
    events**. Use temporal cues like "The video begins with...", "As the
    demonstration progresses...", "Later, the scene shifts to...", and "The
    video concludes by showing..." to create a coherent narrative.
-   Give more descriptive weight to clips with a higher `temporal_ratio`, as
    they represent more of the video's runtime.
-   **Crucially**, also integrate unique, visually striking details from shorter
     clips, especially if they show a distinct action, a close-up of a key
    object, or a significant change in the scene. This ensures the summary is
    both representative and rich in detail.

**Step 3: Refine and Polish.**
-   Ensure the final text is a single, well-written paragraph.
-   Read through your summary one last time to confirm it is 100 percent based
    on visual information and flows logically from beginning to end.

## Handling Inconsistencies & Corner Cases
-   **Conflicting Visuals**: If clips describe conflicting details (e.g., wall
    color changes from 'purple' to 'beige'), either describe the most dominant/
    frequently-seen version or briefly acknowledge the change (e.g., "...in a
    room with walls that appear alternately purple and beige."). **Do not**
    create an inaccurate hybrid description.
-   **Invalid/Empty Clips**: If a clip's caption is an error message, a refusal,
     or contains no visual information, ignore it completely.
```

```
## Output Specification
-   Your output MUST be ONLY a single, valid JSON object.
-   The JSON object must have two keys: `video_id` (string) and `caption` (
    string).
-   The `caption` should be a single paragraph, typically between 30 and 200
    words.

## Input
Video ID: {video_id}
Clips (clip_id  caption  span  ratio):clips
    ## Output:
```

APPENDIX C

SUPPLEMENTARY MATERIAL FOR GPU-ACCELERATED MULTI-RELATIONAL PARALLEL GRAPH RETRIEVAL FOR WEB-SCALE RECOMMENDATIONS

C.1 Pseudo Code of HiPANNS Algorithm

This section presents the pseudocodes of the Hierarchically Parallel Approximate Nearest Neighbor Search (HiPANNS), including inter-candidate HiPANNS (C-HiPANNS) and inter-query (Q-HiPANNS). In Algorithm 1, inter-candidate HiPANNS (C-HiPANNS) allows parallel search within the large-scale graph item base to broaden the global search space for more comprehensive retrieval. In Algorithm 2, inter-query HiPANNS (Q-HiPANNS) adaptively aggregates irregular neural computations for more efficient GPU-accelerated retrieval against launch bound from extreme concurrency.

C.1.1 Implementation Details

We construct HNSW by 4 layers consistently for Video, MVideo, Explore, and Fresh datasets. Detailed statistics of layer-wise node numbers in HNSW structures are presented in Table C.1. Besides, we involve 35 slots of features in user representation learning. The neural metric has about 3 million of learnable parameters. For commercial online serving, we deploy our methods for 2 services with 5 servers, and activate 165 and 119 NVIDIA A30 Tensor Core GPUs, respectively. Detailed GPU equipment of servers is provided in Table C.2.

Algorithm 1: Inter-candidate HiPANNS.

Input: Query user u, HNSW G, Neighbor hops T, heap size K, and number of HNSW layers L.
Output: Top-K retrieved items V_u.

Function `C-HiPANNS(`u,G,T,K,L`)`
  Initialize a K-size max heap Φ_u;
  Initialize K search seeds $S^{(0)} = \{v_1^{(0)}, v_2^{(0)}, \cdots, v_K^{(0)}\}$ and push them into Φ_u;
  for $l = 1, \cdots, L$ **do**
    Initialize search seeds for l-th layer as $S^{(l)} = \Phi_u$;
    foreach $v_i^{(l)} \in S^{(l)}$ *in parallel* **do**
      for $t = 1, \cdots, T$ *in batch* **do**
        Retrieve $v_i^{(l)}$'s t-hop neighbor set $\{v_{i,t_j}^{(l)}\}$;
        Calculate the relevance $\delta_{u,v_{i,t_j}^{(l)}}$ by `Q-HiPANNS(`$\{(u, v_{i,t_j}^{(l)})\}$`)`;
        Push $v_{i,t_j}^{(l)}$ into Φ_u;
  return Φ_u

Algorithm 2: Inter-query HiPANNS

Input: GPU cluster U, U's batch size N_U
Output: Relevance values δ_{B_i}

Function `Q-HiPANNS(`B_i`)`
  Initialize an empty N_U-size batch B_U and current batch length $N'_U = 0$;
  Initialize a queue Ψ;
  Listen to a batch $B_i = \{(u_{i,1}, v_{i,1}), \cdots, (u_{i,N_{B_i}}, v_{i,N_{B_i}})\}$ at real-time and push it into Ψ;
  foreach $B_i \in \Psi$ **do**
    $\epsilon = N_U - N'_U$;
    if $N_{B_i} \leq \epsilon$ **then**
      Concentrate batch as $B_U = B_U \cup B_i$ and $N'_U = N'_U + N_{B_i}$;
      Calculate δ_{B_i} on U;
    else
      Split B_i as $B_i^{[1,\epsilon]}$ and $B_i^{[\epsilon+1,N_{B_i}]}$;
      Concentrate batch as $B_U = B_U \cup B_i^{[1,\epsilon]}$;
      Calculate δ_{B_i} on U;
      Set $B_U = B_i^{[\epsilon+1,N_{B_i}]}$ and $N'_U = N_{B_i} - \epsilon$;
  Concentrate δ_{B_i} if B_i is split;
  return δ_{B_i}

Table C.1: Number of nodes in HNSW layers for four datasets.

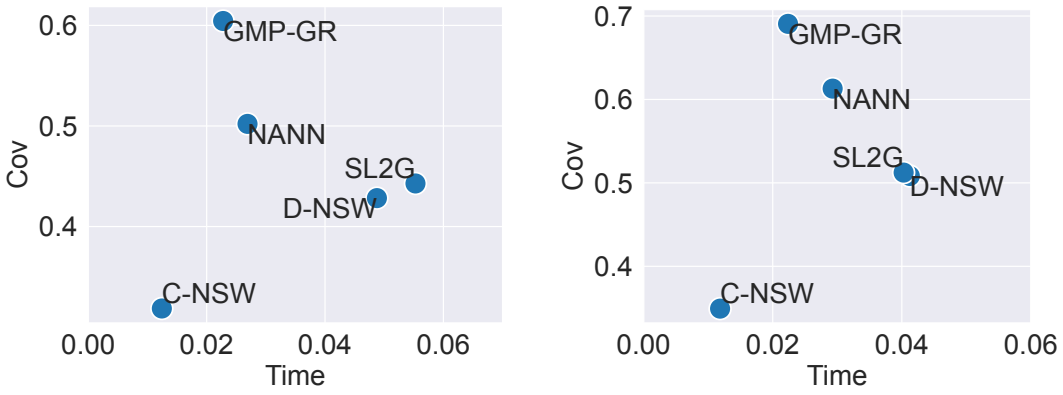
	Video	MVideo	Explore	Fresh
Layer 1	281	214	210	78
Layer 2	4,676	3,560	3,530	1,249
Layer 3	79,249	61,246	60,679	20,874
Layer 4	1,335,473	1,033,857	1,024,531	355,585

Table C.2: GPU equipment of five servers for two services.

	Server 1	Server 2	Server 3	Server 4	Server 5
Service 1	28	24	36	40	37
Service 2	23	15	29	29	23

C.1.2 Additional Efficiency Analysis

Here we analyze the efficiency in terms of inference time and search speed on MVideo and Fresh datasets by scattering results of five methods in Figure C.1.



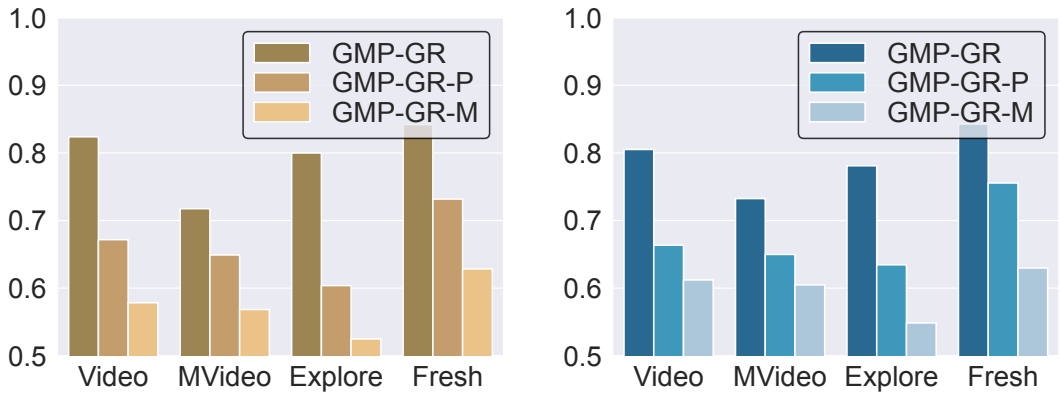
(a) MVideo

(b) Fresh

Figure C.1: Analysis of correlations between retrieval performance and efficiency on four datasets. Methods close to the left top (↖) have better retrieval performance, *i.e.* Cov (↑), with less time costs (←).

C.1.3 Additional Ablation Study

In this subsection, we conduct a further ablation study by comparing GMP-GR with GMP-GR-P and GMP-GR-M according to the evaluated results by Rec@10 and Rec@100 in Figure C.2.



(a) Rec@10

(b) Rec@100

Figure C.2: Additional ablation study of GMP-GR on retrieval performance (Rec@10 and Rec@100).

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